

Meta AD Performance Dashboard

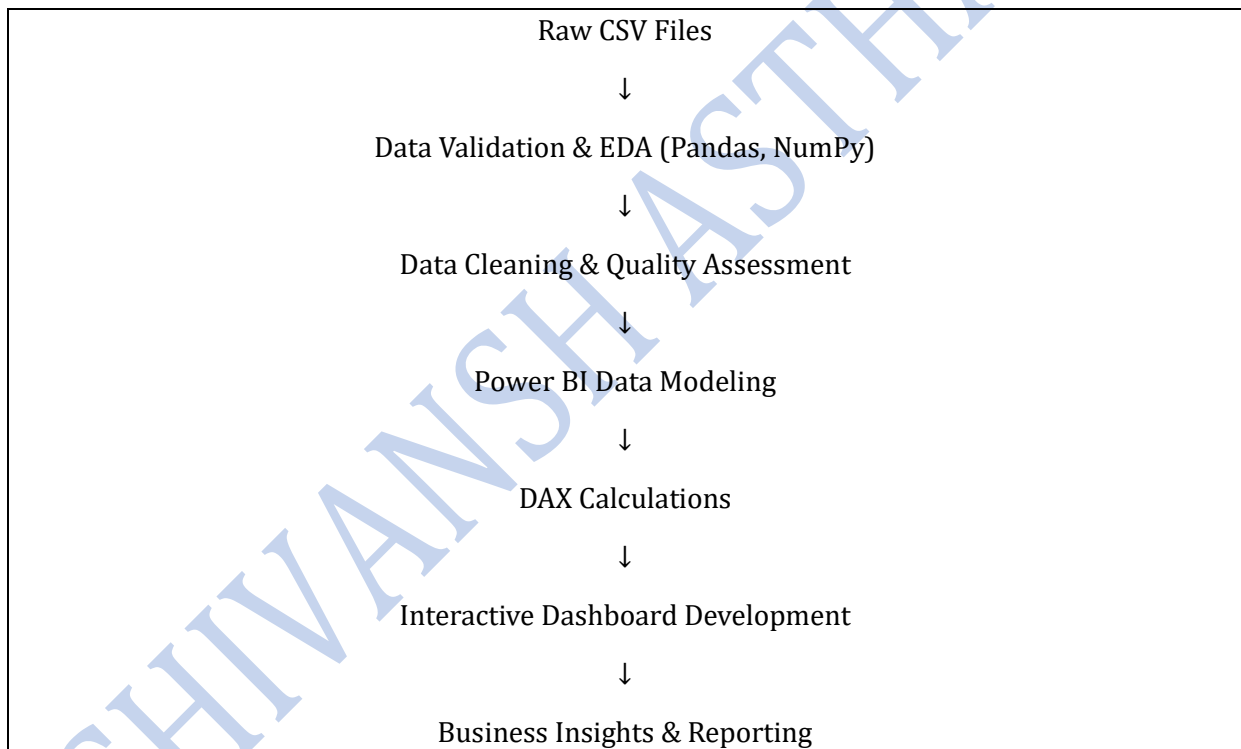
End-to-End Data Analytics Project using Python, Pandas, NumPy & Power BI

Project Overview

The Meta AD Performance Dashboard is an end-to-end Business Intelligence project designed to analyze advertising campaign performance across Meta platforms. The project combines Python-based Exploratory Data Analysis (EDA), data validation, Power BI data modeling, DAX calculations, and interactive dashboard development to transform raw advertising data into actionable business insights.

The primary objective of this project is to help marketing teams monitor campaign performance, understand audience engagement, identify high-performing demographics, and optimize advertising strategies using data-driven decision-making.

Project Workflow



Dataset Description

The project consists of four interconnected datasets.

1. ad_events.csv

Stores user interactions with advertisements.

Examples:

- Impression
- Click
- Like

- Comment
- Purchase
- Share

Purpose

Provides engagement and performance metrics used throughout the dashboard.

2. ads.csv

Contains advertisement-level information.

Examples:

- Ad ID
- Platform Type
- Ad Type
- Campaign Association

Purpose

Used to analyze advertisement performance and platform effectiveness.

3. campaigns.csv

Contains campaign-level information.

Examples:

- Campaign Name
- Campaign Budget
- Start Date
- End Date
- Campaign Duration

Purpose

Used to evaluate campaign performance and budget allocation.

4. users.csv

Contains demographic information.

Examples:

- Country
- Gender
- Age Group
- Interests

Purpose

Used to perform audience segmentation and demographic analysis.

Exploratory Data Analysis (EDA)

Tools Used

import pandas as pd

import numpy as np

The EDA process was performed using Pandas and NumPy to validate data quality and understand the structure of each dataset before importing it into Power BI.

Step 1: Data Loading

Code

```
ad_events = pd.read_csv('ad_events.csv')
```

```
ads = pd.read_csv('ads.csv')
```

```
campaigns = pd.read_csv('campaigns.csv')
```

```
users = pd.read_csv('users.csv')
```

Purpose

To import all datasets into Python DataFrames for analysis.

Output

Four DataFrames were successfully created.

Step 2: Initial Data Inspection

Code

```
df.head()
```

```
df.tail()
```

Performed on:

- ad_events
- ads
- campaigns
- users

Purpose

To inspect the structure and formatting of the datasets.

Outcome

Verified:

- Column names
- Record formatting

- Data consistency
-

Step 3: Dataset Structure Analysis

Code

```
df.info()
```

Purpose

To examine:

- Number of rows
- Number of columns
- Data types
- Non-null values

Outcome

Confirmed that dataset schemas were correctly imported and suitable for Power BI modeling.

Step 4: Statistical Summary

Code

```
df.describe()
```

Purpose

To generate descriptive statistics for numerical columns.

Outcome

Analyzed:

- Mean
- Minimum
- Maximum
- Standard Deviation
- Quartiles

This helped understand data distribution and identify unusual values.

Step 5: Missing Value Analysis

Code

```
df.isnull().sum()
```

Purpose

To identify incomplete records.

Outcome

All datasets were evaluated for missing values.

This validation ensured accurate KPI calculations and dashboard reporting.

Step 6: Dataset Dimension Analysis

Code

```
df.shape
```

Purpose

To determine dataset size.

Outcome

Provided row and column counts for each dataset, helping estimate model complexity and dashboard performance.

Step 7: Duplicate Record Validation

Code

```
df.duplicated().sum()
```

Purpose

To identify duplicate records.

Outcome

Verified data integrity and prevented inflation of campaign and engagement metrics.

EDA Conclusion

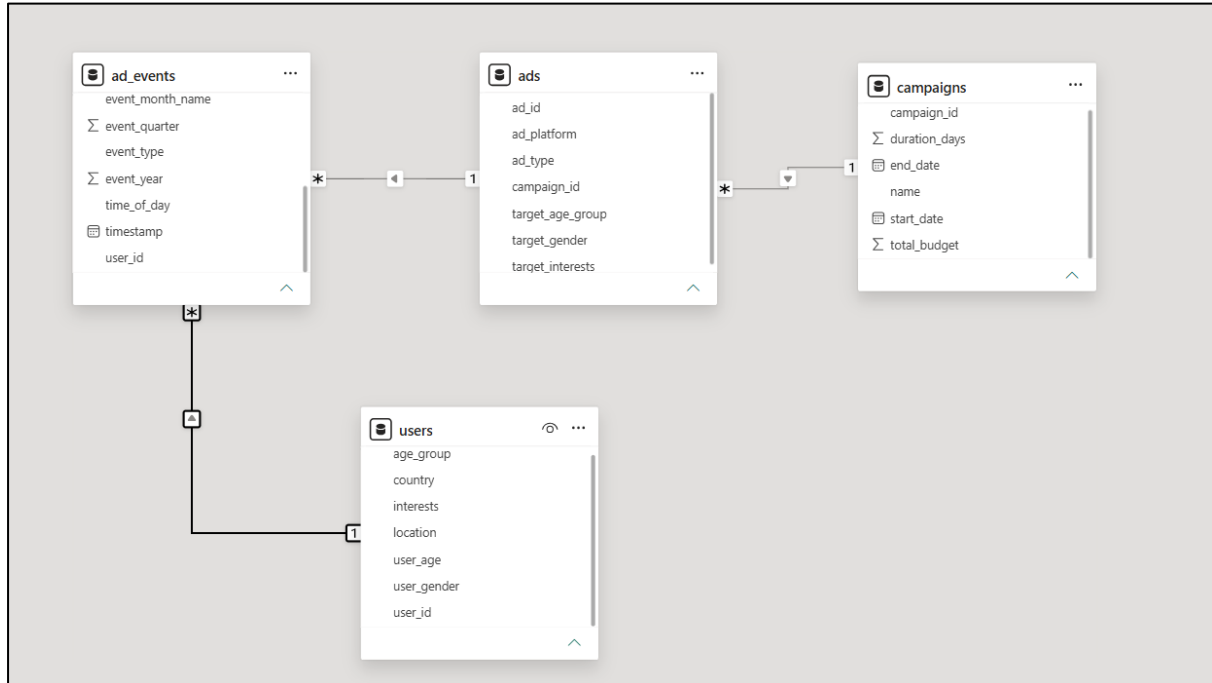
- ✓ The EDA process confirmed:
- ✓ Successful data loading
- ✓ Valid dataset structures
- ✓ Consistent data types
- ✓ Missing value assessment completed
- ✓ Duplicate validation completed
- ✓ Numerical distributions analysed

The datasets were determined to be suitable for business intelligence reporting and dashboard development.

Power BI Data Modeling

After completing EDA, the datasets were imported into Power BI and connected through relationships.

Data Model Structure



Modeling Objectives

- Establish relationships between campaigns and ads.
- Connect user interactions to demographic data.
- Enable cross-filtering across visualizations.
- Support DAX calculations and KPIs.

Power Query Transformations

The following transformations were performed inside Power BI:

Date Transformation

- Converted date columns into proper date format.
- Extracted month information for trend analysis.

Data Type Validation

- Verified numerical, categorical, and date fields.

Relationship Preparation

- Ensured key columns were suitable for relationship creation.

DAX Measures and KPIs

Several measures were created to support dashboard reporting.

Total Users

```
DISTINCTCOUNT(users[user_id])
```

Total Events

```
COUNTROWS(ad_events)
```

Total Impressions

```
CALCULATE(  
    COUNTROWS(ad_events),  
    ad_events[event_type] = "Impression"  
)
```

Total Clicks

```
CALCULATE(  
    COUNTROWS(ad_events),  
    ad_events[event_type] = "Click"  
)
```

Engagement Rate

$$\frac{(\text{Clicks} + \text{Likes} + \text{Comments} + \text{Shares} + \text{Purchases})}{\text{Impressions}}$$

Dashboard Overview

The final dashboard provides a centralized view of advertising performance, audience behavior, and engagement metrics.

Interactive Filters

The dashboard includes dynamic slicers for:

Interest

Allows audience segmentation based on interests.

Platform Type

Enables platform-level analysis.

Platforms:

- Facebook

- Instagram

Gender

Supports demographic segmentation.

Options:

- Male
- Female
- Other

KPI Summary

KPI	Value
Total Users	10K
Total Impressions	340K
Total Budget	\$2.54M
Total Events	400K
Total Campaigns	50
Total Clicks	40K
Engagement Rate	15.91%
Total Ads	200
Total Likes	12K
Total Shares	2K

Dashboard Visual Analysis

Top 5 Countries by Total Events

Purpose: To identify regions generating the highest engagement.

Key Findings

Country	Events
United States	115K
United Kingdom	57K
India	36K
Mexico	19K
Japan	19K

Business Insight

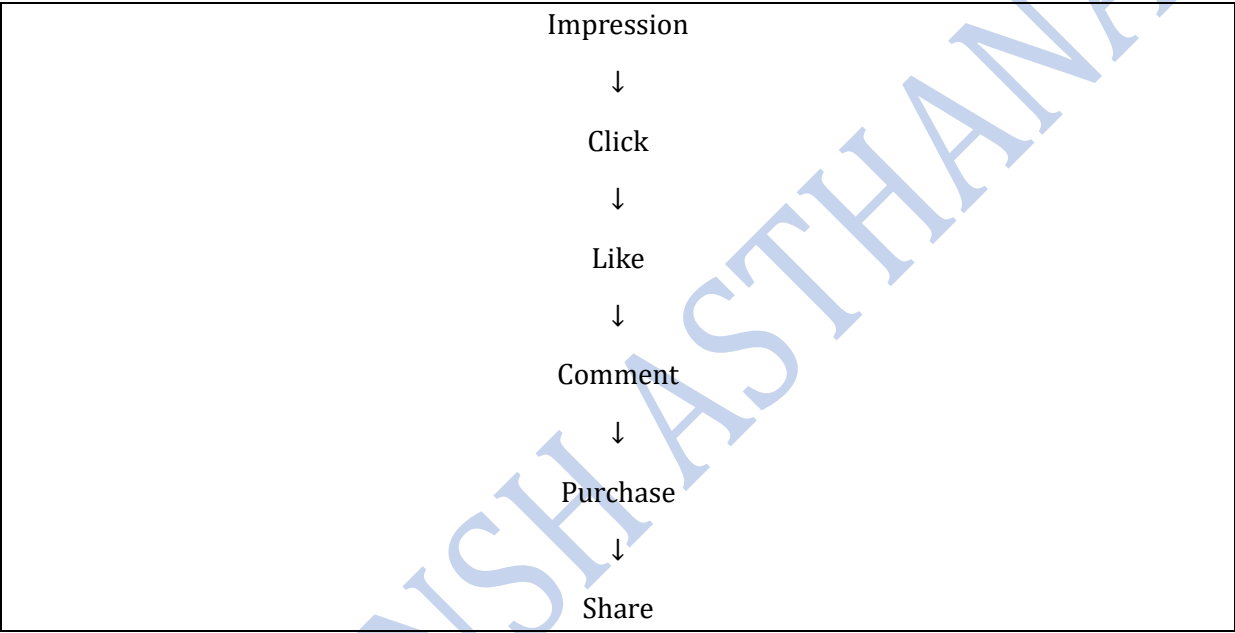
The United States is the strongest-performing market, while India demonstrates significant growth potential.

Event Type Funnel

Purpose

To analyze user movement through the advertising engagement funnel.

Funnel Flow



Key Findings

Event Type	Count
Impression	340K
Click	40K
Like	12K
Comment	4K
Purchase	2K
Share	2K

Business Insight

A significant drop-off exists between impressions and deeper engagement actions, highlighting opportunities for conversion optimization.

Monthly Events Trend

Purpose

To monitor engagement performance over time.

Key Findings

- Events increased steadily from May to July.
- July recorded the highest engagement.
- August showed a sharp decline.

Business Insight

The trend helps identify seasonal effects, campaign timing issues, and optimization opportunities.

Audience Engagement by Age Group

Purpose

To identify the most responsive demographic segments.

Key Findings

- Age group 35–44 generated the highest engagement.
- Age group 18–24 closely followed.
- Age group 25–34 showed comparatively lower engagement.

Business Insight

Future campaigns can prioritize higher-performing audience segments to maximize engagement.

Business Impact

The dashboard enables stakeholders to:

- Monitor advertising performance.
 - Track audience engagement.
 - Evaluate campaign effectiveness.
 - Identify high-performing geographic regions.
 - Analyze user demographics.
 - Optimize budget allocation.
 - Improve targeting strategies.
 - Support data-driven marketing decisions.
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Technologies Used

Programming & Analysis

- Python

- Pandas
- NumPy

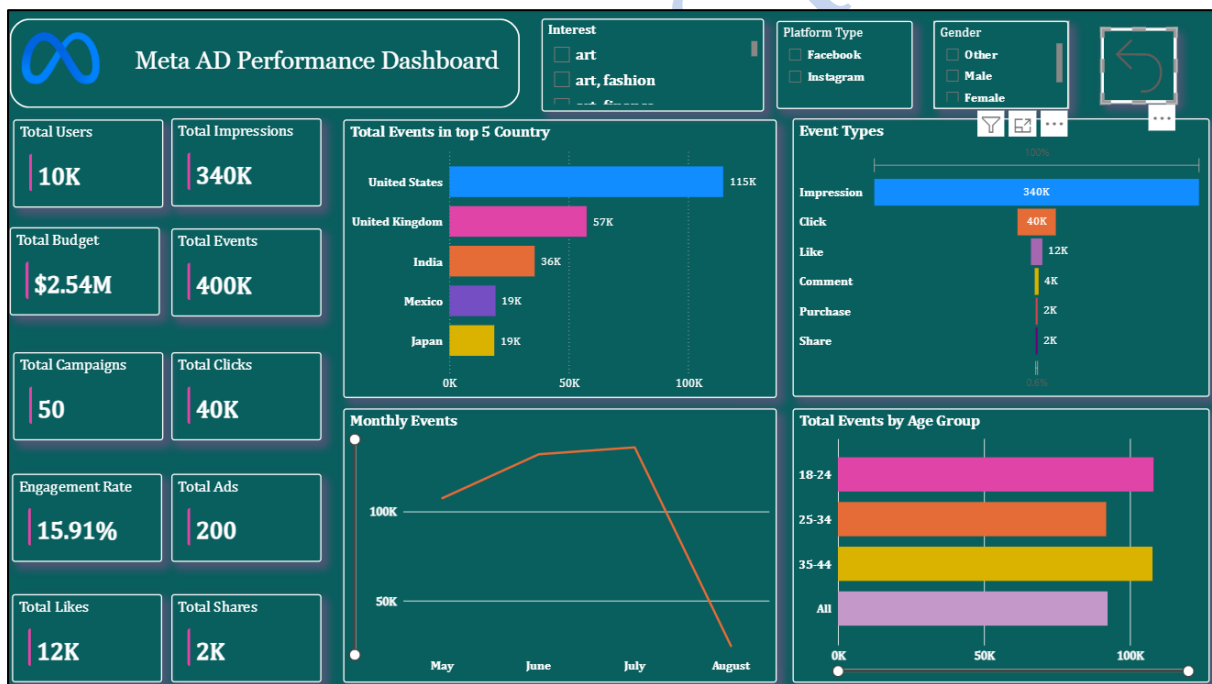
Business Intelligence

- Power BI
- Power Query
- DAX

Data Modeling

- Star Schema Relationships
- Data Validation
- KPI Design
- Interactive Filtering

DASHBOARD



Conclusion

This project demonstrates a complete Business Intelligence workflow, starting from raw advertising datasets and ending with an interactive Power BI dashboard. Through Python-based EDA, Power BI data modeling, DAX calculations, and business-focused visualizations, the project successfully transforms raw marketing data into actionable insights that support strategic decision-making and campaign optimization.